AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 8 · STUDENT LESSON

Loss functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Does knowing a network output tell us whether it is accurate?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Calculate per-example squared error and mean squared error. Try the recall before opening notes.

Your learning target

I can calculate per-example **squared error** and **mean squared error**.

I can distinguish **residual** cancellation from a low loss.

I can separate **training** loss from independent evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate per-example **squared error** and **mean squared error**

Distinguish **residual** cancellation from a low loss

Separate **training** loss from independent evaluation

Retrieve what you already know

> Does knowing a network **output** tell us whether it is accurate?

My starting explanation

This week's end goal

By the end, I can explain loss functions using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Loss functions case. Separate observations, given facts and assumptions before choosing an explanation.

A loss assigns a numerical penalty to a **prediction** compared with a **target**. For this regression lesson, **residual** e = **prediction** - **target**, and **squared error** is e squared.

Squaring makes both positive and negative **residual**s contribute nonnegative penalties. We use ordinary **mean squared error** without a one-half factor.

Example · Target · Prediction · Residual · Squared error

> **Example:** A
>
> **Target:** 1
>
> **Prediction:** 2
>
> **Residual:** 1
>
> **Squared error:** 1

> **Example:** B
>
> **Target:** 3
>
> **Prediction:** 2
>
> **Residual:** -1
>
> **Squared error:** 1

> **Example:** C
>
> **Target:** 2
>
> **Prediction:** 4
>
> **Residual:** 2
>
> **Squared error:** 4

The **mean squared error**, or MSE, is (1 + 1 + 4) / 3 = 2. It summarises these three examples, not every possible future **input**. If **target**s use cm, **residual**s use cm and MSE uses square cm.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Loss functions. Connect each labelled input, deciding step and result.

For **target**s [1, 3] and **prediction**s [2, 2], the mean **residual** is zero because 1 and -1 cancel. MSE is 1, correctly recording that both **prediction**s miss.

A mean **residual** can describe directional tendency, but it is not a substitute for an error magnitude.

```
def mse(predictions, targets):
    if not targets or len(predictions) != len(targets):
        raise ValueError("Need equal nonempty sequences")
    errors = [(p - t) ** 2
              for p, t in zip(predictions, targets)]
    return sum(errors) / len(errors)

print(mse([2, 2, 4], [1, 3, 2]))  # 2.0
print(mse([2, 2], [1, 3]))        # 1.0
```

The guard prevents silently ignoring an unmatched example or dividing by zero. It still assumes paired, numeric, valid records. Wrong row alignment produces a meaningless comparison even when list lengths agree.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Loss functions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Targets [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Choose MAE instead of MSE as the sole criterion.

Predicted result

Prediction: A and B tie at MAE1.

Why the result follows

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Loss functions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Target	The recorded value a model is intended to predict
Residual	Prediction minus target under the sign convention used here
Squared error	The square of a numerical residual
Mean squared error	The average squared error over the evaluated examples

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Loss functions, predict a step, then compare with its explanation.

Use the same targets and denominator

CANDIDATE PREDICTIONS	SQUARED ERRORS	MSE
P is [2, 2, 4]	[1, 1, 4]	6 divided by 3 is 2
Q is [1, 2, 2]	[0, 1, 0]	1 divided by 3
Baseline is [2, 2, 2]	[1, 1, 0]	2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Loss functions

Compare candidate losses with the same **target**s and denominator.

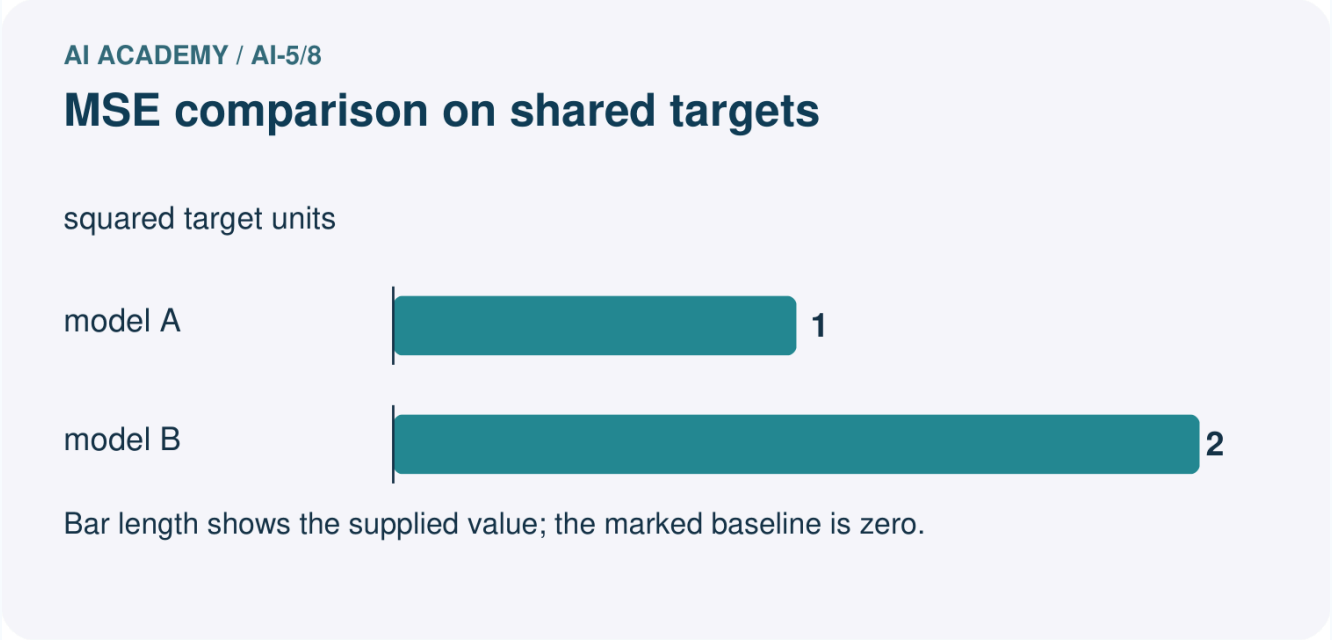
The facts we will use

Supplied facts

Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/8

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A squared errors

A residuals [-1, 1] give squares [1, 1], MSE1.

↓

2. B squared errors

B residuals [0, 2] give squares [0, 4], MSE2.

↓

3. Compare objective

A wins under MSE; both MAEs equal1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A squared errors	A residuals [-1, 1] give squares [1, 1], MSE1.
2. B squared errors	B residuals [0, 2] give squares [0, 4], MSE2.
3. Compare objective	A wins under MSE; both MAEs equal1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

What this example does not establish

Keep the stated limit

Training loss is not automatically a held-out performance estimate or a complete measure

of use consequences.

Keep the stated limit

Read the labelled picture: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Choose MAE instead of MSE as the sole criterion. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Loss functions claim and explain a repair using the supplied evidence.

Check the claim against the facts

Squaring gives larger **residual**s more influence: **residual** 4 contributes 16, four times the penalty of **residual** 2. This can be useful when large misses matter, but it can also make outliers influential.

Mean absolute error uses absolute **residual**s instead; it can rank candidates differently.

MSE is appropriate here because **target**s are numerical quantities. A class identifier such as 0, 1 or 2 may merely name a category; its numeric gap does not automatically measure error severity.

Later classification models need a loss suited to their **output**s and task.

Locate the earliest unsupported step

A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Loss functions with support, then explain the deciding step in your own words.

For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.

Explain your prediction to a partner before comparing results.

Practise with less help: Expose cancellation

Use workbook W2 for the connected example “Loss functions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Loss functions. Proposed explanation: “Average **residual** is the loss”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Loss functions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Targets [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

```
def mse(predictions, targets):
    if not targets or len(predictions) != len(targets):
        raise ValueError("Need equal nonempty sequences")
    errors = [(p - t) ** 2
              for p, t in zip(predictions, targets)]
    return sum(errors) / len(errors)

print(mse([2, 2, 4], [1, 3, 2]))  # 2.0
print(mse([2, 2], [1, 3]))        # 1.0
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Loss functions. Explain why each step is justified.

Explain → apply → challenge

For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

> A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled loss functions **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate per-example **squared error** and **mean squared error**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Training loss measures the cases used to adjust parameters. Validation loss can guide choices, so repeated use makes it development **evidence**.

A reserved final test supports a separate evaluation claim only while it remains outside those choices. Always report the split, number of cases and loss definition.

A model reports MSE zero on its training rows. What has this shown, and why is it insufficient to claim perfect future predictions?

Next week studies local changes in a loss, preparing us to choose a direction for a parameter update.

A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Loss functions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Use the same targets and denominator

CANDIDATE PREDICTIONS	SQUARED ERRORS	MSE
P is [2, 2, 4]	[1, 1, 4]	6 divided by 3 is 2
Q is [1, 2, 2]	[0, 1, 0]	1 divided by 3
Baseline is [2, 2, 2]	[1, 1, 0]	2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare an unfamiliar evaluation independently

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Loss functions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compare candidate losses with the same **target**s and denominator.

By the end, I can	How I will show it
Calculate per-example squared error and mean squared error	Explain the deciding step, show a trace or test, and state one limit.
Distinguish residual cancellation from a low loss	Explain the deciding step, show a trace or test, and state one limit.
Separate training loss from independent evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Loss functions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate per-example **squared error** and **mean squared error**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare candidate losses with the same **target**s and denominator.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A wins under MSE; both MAEs equal1.

1. Compare objective

2. A **squared error**s

3. B **squared error**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare objective A wins under MSE; both MAEs equal1. For this loss calculation, the signed error is **prediction** minus **target**.

MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more.

State the objective before selecting a model, because these two objectives give different comparisons.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Choose MAE instead of MSE as the sole criterion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. A and B tie at MAE1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A and B tie at MAE1. **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Repair a missing building block

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

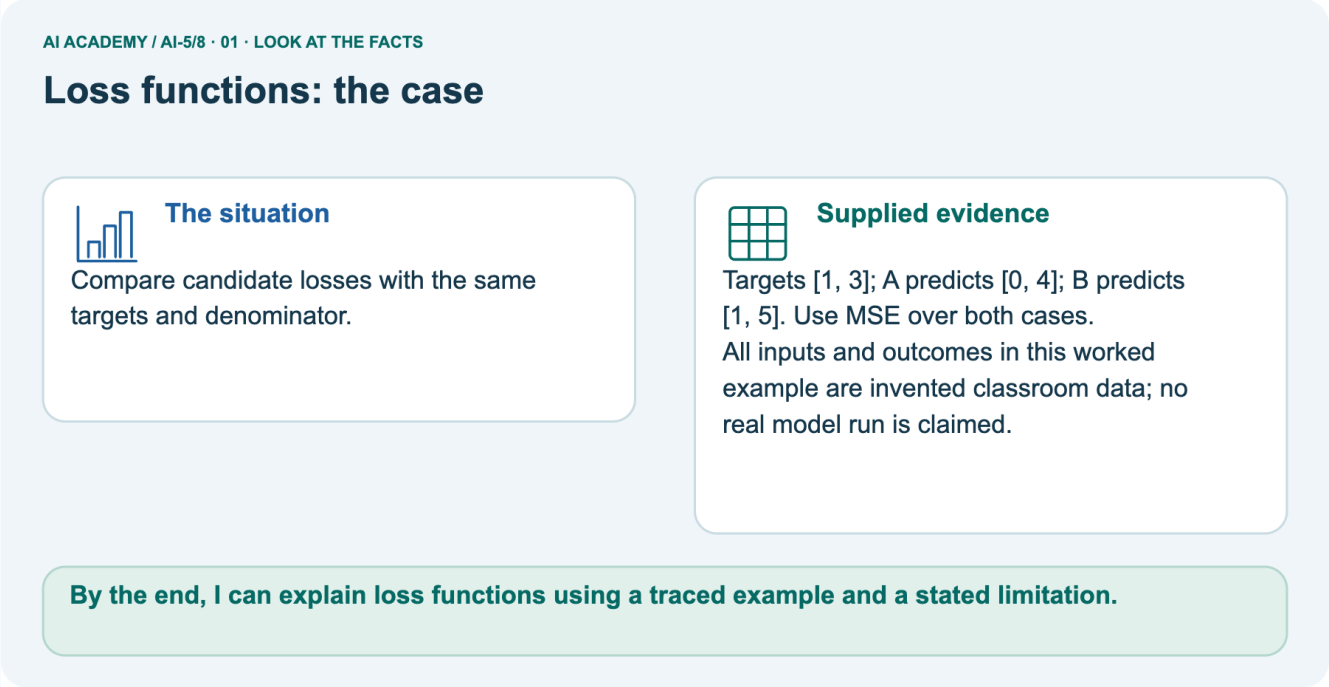
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Loss functions graphic using its labels, deciding evidence and relevant limitation.

01. Loss functions: the case



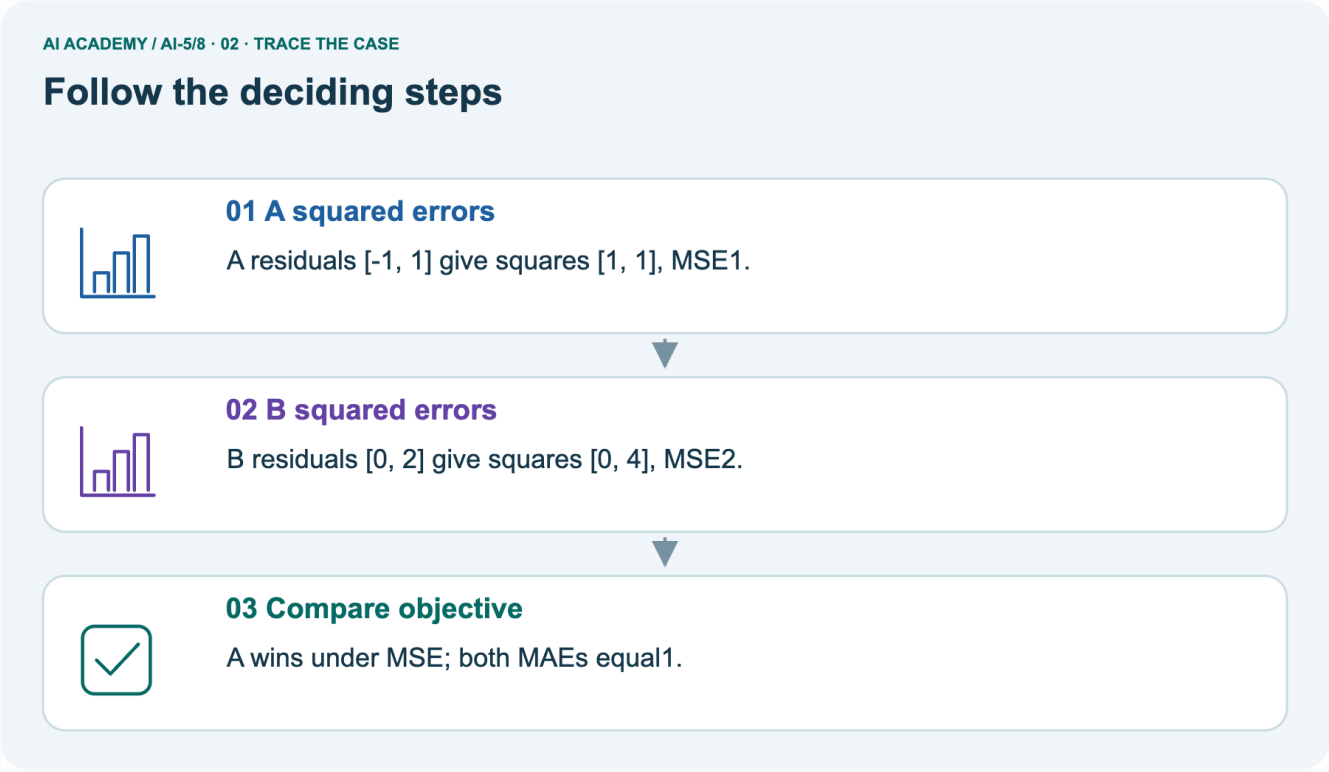
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare candidate losses with the same **target**s and denominator. Supplied facts: **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

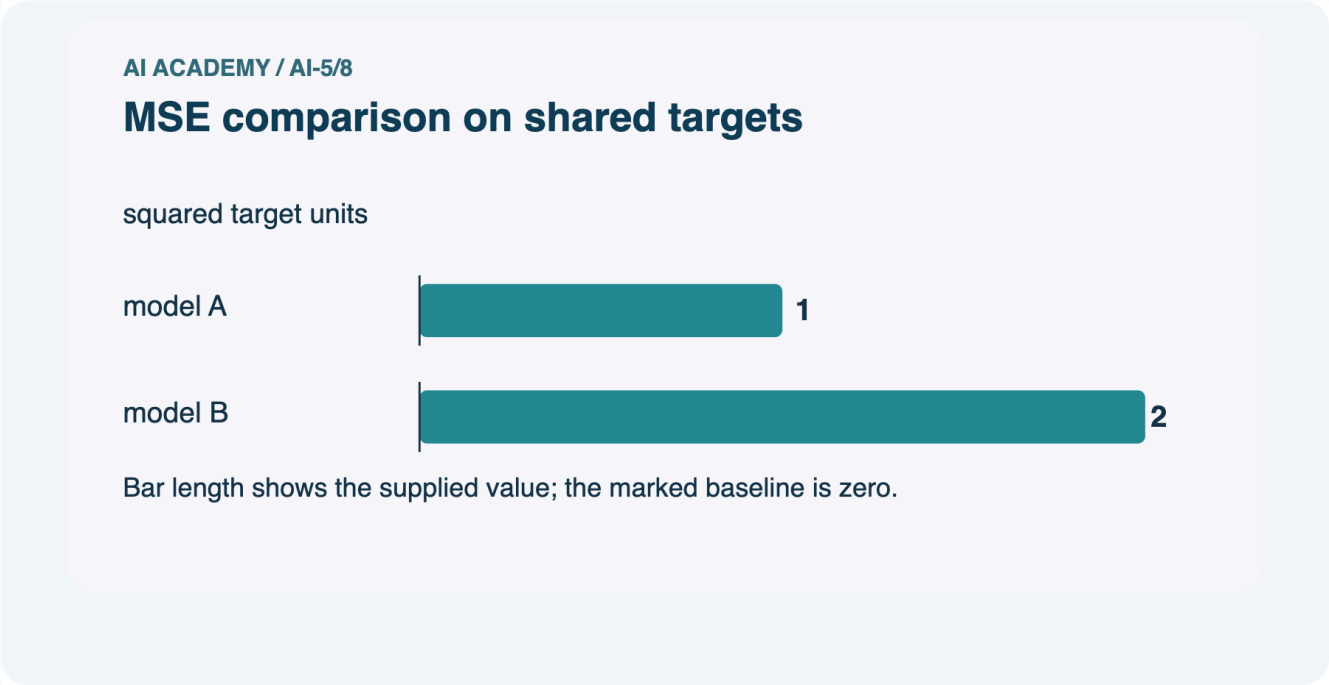


Follow the labelled picture with the matching explanation. Open the vector diagram

A **squared error**s: A **residual**s [-1, 1] give squares [1, 1], MSE1. → B **squared error**s: B **residual**s [0, 2] give squares [0, 4], MSE2. → Compare objective: A wins under MSE; both MAEs equal1.

> Point to each numbered stage and explain which recorded **input** it uses.

03. MSE comparison on shared targets

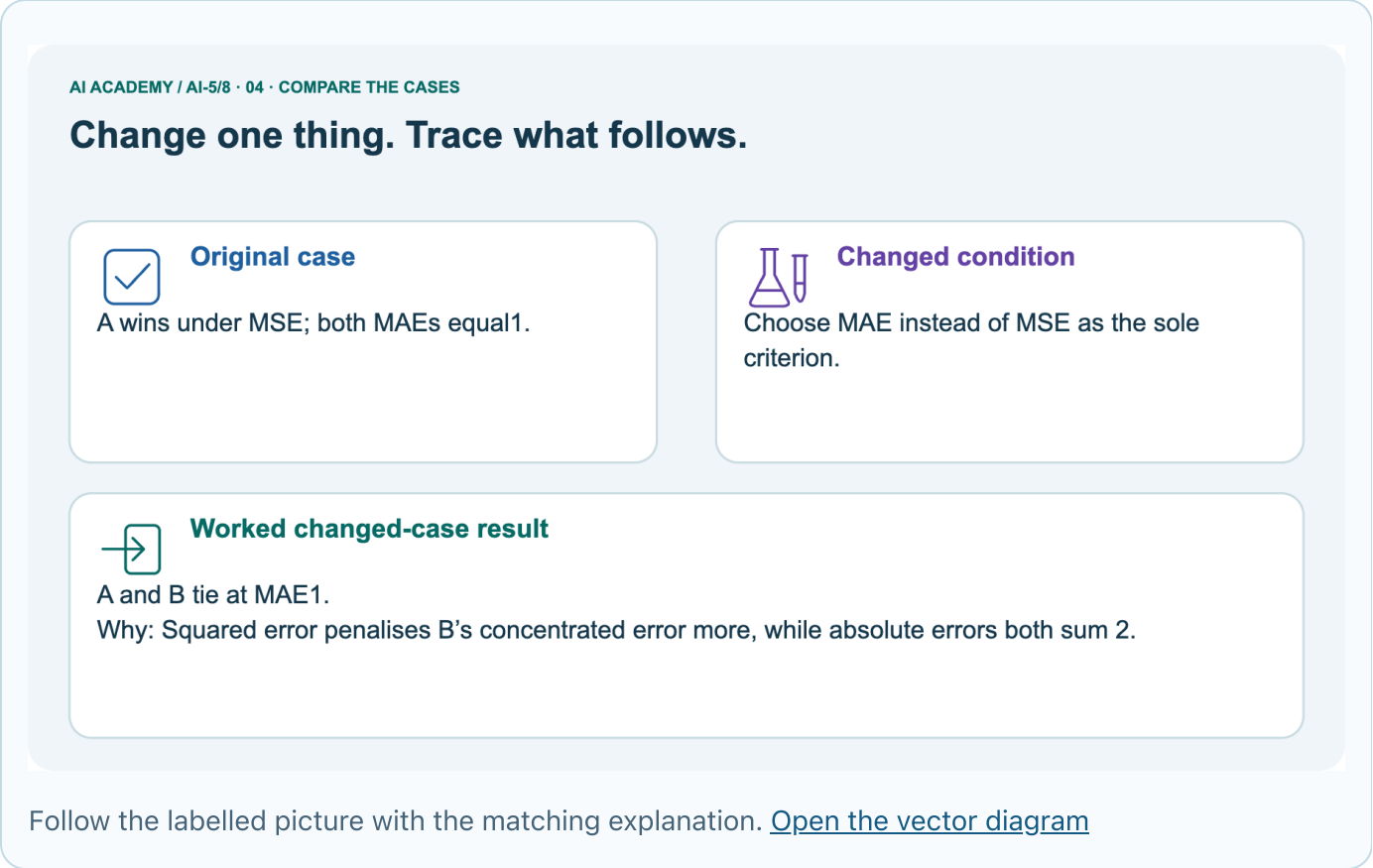


Follow the labelled picture with the matching explanation. Open the vector diagram

MSE comparison on shared **target**s. Units: squared **target** units. model A: 1. model B: 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



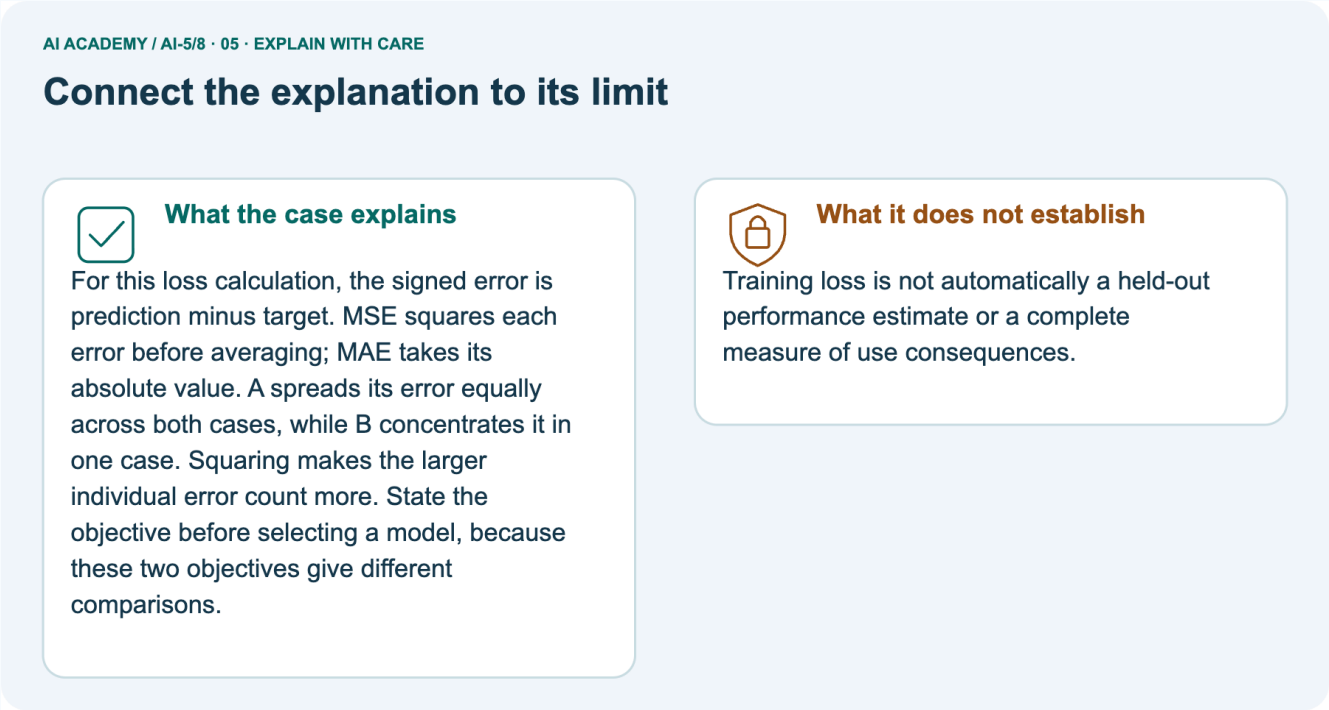
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A wins under MSE; both MAEs equal1. Changed condition: Choose MAE instead of MSE as the sole criterion. Expected result: A and B tie at MAE1.

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

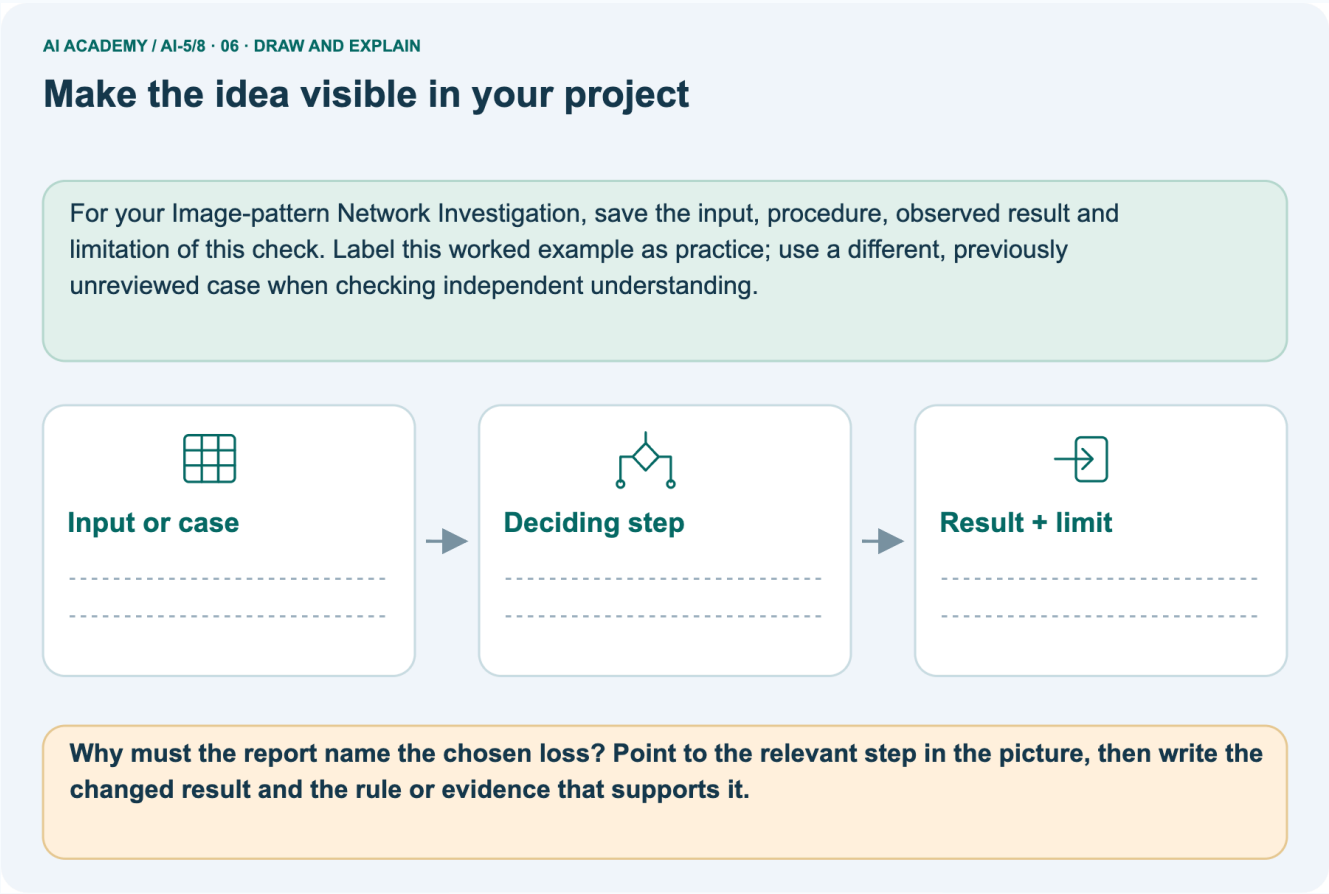
For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Limit: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.